Introduction

Real survival studies routinely include covariates whose value changes during follow-up: treatment initiation, laboratory measurements repeated at each visit, secondary events that alter prognosis. Including these as time-fixed baseline values misrepresents the exposure history; pretending the baseline value applied throughout follow-up biases hazard-ratio estimates and is a leading cause of immortal-time errors. The standard remedy is the counting-process data layout, which splits each subject's follow-up into intervals during which covariate values are constant and lets the partial likelihood combine the contributions correctly.

Prerequisites

A working understanding of Cox regression, the partial likelihood, and the distinction between baseline (time-fixed) and time-varying covariates.

Theory

In counting-process format each row represents one interval $(\tau_{ij}^{\text{start}}, \tau_{ij}^{\text{stop}}]$ during which covariate values for subject i are constant. The columns are:

- `tstart`: interval start.
- `tstop`: interval end.
- `event`: 1 if the event occurred at `tstop`, otherwise 0.
- Covariates with their values during the interval.

The partial likelihood evaluates each subject's risk-set contribution using whichever interval includes the event time, so each event time naturally compares subjects with their then-current covariate values. Robust ("sandwich") variance estimators, requested via `cluster = id`, account for the within-subject correlation induced by multiple rows per subject.

Assumptions

Covariate values are constant within each interval and accurately measured. The covariate is "internal" if its trajectory could be affected by the event of interest (e.g., a biomarker that changes after treatment), or "external" if it is determined exogenously (e.g., scheduled treatment dose); inference from internal covariates requires careful causal interpretation.

R Implementation

```r
library(survival)

data(cgd, package = "survival")
head(cgd)

# Multi-row-per-subject: counting-process format
fit <- coxph(Surv(tstart, tstop, status) ~ treat + inherit,
             data = cgd, cluster = id)
summary(fit)
```

Output & Results

`summary(fit)` produces the usual Cox-regression output with hazard ratios, confidence intervals, and Wald tests for each covariate, plus robust standard errors that respect the within-subject clustering. The number of subjects (one per cluster) and the number of events are reported alongside the number of intervals.

Interpretation

A reporting sentence: "After accounting for the time-varying treatment status across each subject's intervals, prophylactic treatment reduced the hazard of CGD infection by 60 % (HR 0.40, 95 % CI 0.25 to 0.63); robust standard errors were clustered on subject identifier to account for repeated intervals." When the time-varying covariate is treatment status, always describe whether subjects could move between exposure groups and how often.

Practical Tips

- Use `survSplit()` (or `tmerge()` for multi-event data) to convert wide-format clinical data into the counting-process layout; doing this by hand is error-prone.
- Always specify `cluster = id` (or `cluster(id)` in the formula) so robust standard errors account for multiple rows per subject; ignoring clustering produces SEs that are too small.
- Check for overlapping intervals: the same subject should never have two rows whose intervals share any time.
- For external time-varying covariates (treatment delivered at fixed planned times), landmark analysis is a cleaner alternative that avoids the data-management overhead.
- For internal time-varying covariates (biomarkers that respond to the disease course), joint longitudinal-survival models (`JM`, `joineRML`) are preferable because they correctly handle the feedback loop.
- Be explicit in the methods section about which covariates were treated as time-varying and how often they were updated; readers cannot reproduce the analysis without this information.

R Packages Used

`survival` for `coxph()` with counting-process input, `survSplit()`, and `tmerge()`; `dynpred` for landmark and dynamic-prediction alternatives; `JM` and `joineRML` for joint longitudinal-survival models; `mstate` for multi-state extensions when intervals correspond to state transitions.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Survival primer: KM, log-rank, Cox PH
- Time-varying covariates and immortal-time bias
- Competing risks and multistate models
- Survival ML (random survival forest, DeepSurv)
- Time-dependent Brier, IPA, external validation